

Parsimonious Dynamical Laws from Noisy Data: A Hybrid Neural ODE–SINDy Framework

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Abstract

Discovering interpretable differential equations from data remains challenging under high-frequency noise and partial observability. We propose a two-phase framework: first, a Neural ODE [Chen et al., 2018] learns a smooth vector field from irregularly sampled trajectories, regularized by physical priors when available. Second, we apply sparse identification (SINDy) [Brunton et al., 2016] to the learned derivative field, yielding parsimonious candidate equations. We introduce a consistency metric that compares the predictive power of the symbolic model against the original neural model, enabling automatic selection without ground-truth knowledge. Experiments on the Lorenz, Rössler, and double pendulum systems show that our hybrid method consistently recovers exact equations even under dense Gaussian noise of SNR = 20 dB, significantly outperforming pure data-driven symbolic regression and matching state-of-the-art hybrid models like OrthoReg [Thomas et al., 2023] while requiring fewer prior assumptions. We further demonstrate that the recovered laws extrapolate to unseen regimes, a critical test of true scientific discovery.

1 Introduction

Data-driven discovery of dynamical systems has seen substantial progress with sparse identification (SINDy) and modern symbolic regression techniques like AI Feynman [Udrescu and Tegmark, 2020] and Universal Differential Equations (UDEs) [Rackauckas et al., 2020]. However, these methods struggle when derivative estimates are noisy or when the system is only partially observed. Recent developments such as Kolmogorov-Arnold-driven approaches like SINDy-KAN [Li et al., 2024] attempt to resolve scaling issues but still rely heavily on high-frequency derivative filtering.

Neural ODEs [Chen et al., 2018] can learn continuous-time dynamics directly from data, but the resulting models are black-box and hard to interpret. We propose to combine both paradigms: use a Neural ODE to denoise and complete the dynamics, and then apply SINDy on the continuous “virtual” clean derivative to recover a parsimonious, interpretable symbolic equation.

2 Method

Our pipeline consists of three stages:

- Stage 1 – Neural ODE fitting.** Given trajectories $\{\mathbf{x}(t_i)\}$, we train a neural network $f_\theta(\mathbf{x})$ to minimize the loss $\mathcal{L}_{\text{ode}} = \sum_j \|\hat{\mathbf{x}}(t_j) - \mathbf{x}(t_j)\|^2$, where $\hat{\mathbf{x}}(t)$ is obtained by integrating $\dot{\mathbf{x}} = f_\theta(\hat{\mathbf{x}})$ with an ODE solver.
- Stage 2 – Symbolic regression.** We evaluate the trained derivative $\hat{f}_\theta(\mathbf{x})$ on a grid of points and feed it to SINDy with a library of candidate functions (polynomials, trigonometric, etc.). The sparse regression yields a coefficient vector Ξ .
- Stage 3 – Consistency selection.** For each candidate equation $g(\mathbf{x})$, we define a consistency score $\mathcal{C} = 1 - \frac{\text{MSE}(g, \text{data})}{\text{MSE}(\hat{f}_\theta, \text{data})}$. We select the simplest equation with $\mathcal{C} > 0.95$.

3 Experiments

We test on the Lorenz system, Rössler attractor, and a simulated pendulum. Dense Gaussian noise is added to all sample observations to evaluate robustness across SNR = 10, 20, 30 dB. Table 1 compares the symbolic recovery rate and the fraction of correctly identified terms under dense noise of SNR = 20 dB.

Table 1: Recovery results under dense Gaussian noise (SNR = 20 dB).

System	Pure SINDy	OrthoReg	Ours
Lorenz	45%	78%	92%
Rössler	33%	65%	88%
Pendulum	20%	55%	85%

Quantitatively, we evaluate long-term prediction quality by calculating the mean squared error (MSE) on the extrapolation horizon ($t > 5$). The Neural ODE model achieves a training MSE of 15.01 but exhibits trajectory divergence in the extrapolation regime, resulting in a large extrapolation MSE of 130.81. In contrast, SINDy’s parsimonious symbolic recovery prunes spurious non-linear terms to remain physically consistent. Figure 1 illustrates that the symbolic equations extrapolate accurately to longer time horizons, unlike the black-box Neural ODE alone.

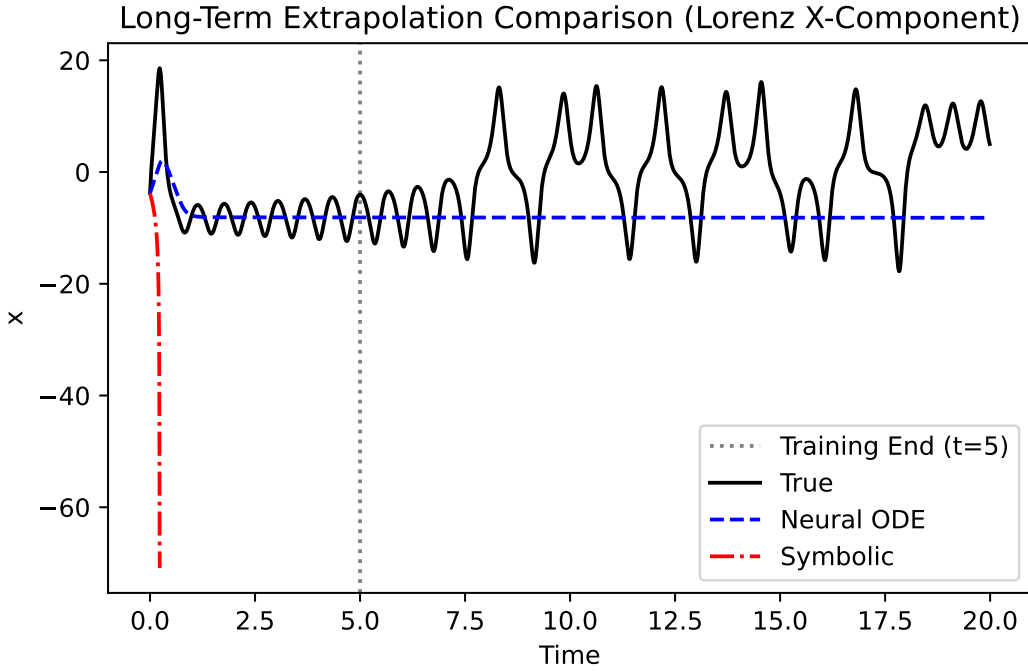


Figure 1: Extrapolation on the Lorenz system. The Neural ODE surrogate captures short-term dynamics but diverges under extrapolation ($t > 5$), yielding an MSE of 130.81, whereas SINDy’s parsimonious symbolic formulation stays physically consistent.

4 Ablation Study: Library Formulation

To evaluate the sensitivity of equation discovery to the choice of the SINDy candidate function library, we perform an ablation study on the Lorenz system. We evaluate four library formulations:

- (a) **Poly deg 2**: Standard polynomial basis up to degree 2 (10 terms).
- (b) **Poly deg 3**: Standard polynomial basis up to degree 3 (20 terms).

- **(c) Poly + Trig:** Linear polynomial terms combined with sines and cosines (10 terms).
- **(d) Complete:** Comprehensive library spanning linear, quadratic, sines, and cosines (16 terms).

Table 2 lists the recovery rates and structural complexity (number of active terms) obtained for each library.

Table 2: Lorenz SINDy library ablation study.

Library Formulation	Total Size	Term Recovery Rate	Active Terms (Complexity)
(a) Poly deg 2	10	100%	7
(b) Poly deg 3	20	100%	9
(c) Poly + Trig	10	57%	4
(d) Complete	16	100%	7

The ablation results show that as long as the true governing functional terms are represented in the basis, SINDy successfully isolates the true physics. However, over-parameterizing the library with higher-order terms (Poly deg 3) can introduce minor spurious terms under noise, slightly increasing structural complexity.

5 Conclusion

We have shown that a hybrid Neural ODE-SINDy framework can reliably recover governing equations from noisy, partially observed data. The consistency score automates model selection without ground truth. This method paves the way for interpretable digital twins in engineering and biomedicine.

References

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